

Tensor Omics: Core Preprocessing, Analysis Pipeline, and Geometric Tools for Semantic Interpretation

Asis Hallab, “el Bobito”

April 13, 2025

1 Overview

This document outlines the sequential pipeline used in Tensor Omics for preprocessing, vector space construction, outlier detection, and geometric analysis of gene expression data.

2 From Raw Data to Expression Vectors

The starting point for all downstream analyses is a table of gene-wise expression measurements across a set of biological samples, such as tissues, developmental stages, or experimental conditions. These measurements may originate from transcriptomic, proteomic, or metabolomic experiments, typically normalized to account for sequencing depth and technical variation.

What is Gene Expression Data? For each gene, a real-valued quantity is recorded that reflects its biological activity in a given sample. In the case of transcriptomics, this is often the abundance of messenger RNA (mRNA) molecules transcribed from that gene, estimated via sequencing and mapped back to the gene’s known sequence. The result is a non-negative real number proportional to the gene’s transcriptional activity in that sample. After normalization and transformation, these values can be treated as coordinates in a real vector space.

Thus, each gene is represented as a vector $\vec{v} \in \mathbb{R}^n$, where n is the number of biological samples (e.g., tissues or time points), and each component $v_i \geq 0$ corresponds to the expression level in sample i . These vectors encode the distribution of gene activity across conditions and form the geometric basis for all downstream analyses.

Note on Stress Responses. In certain studies, a subset of the samples may correspond to stress conditions. Here, “stress” refers to any external or internal perturbation—such as mechanical damage, heat shock, pathogen infection, or nutrient deprivation—that triggers measurable changes in gene expression. These are often analyzed relative to baseline conditions using log fold-change transformations, such that the resulting vectors reflect differential expression patterns, including up- and downregulation across dimensions.

3 Normalization of Expression Values

Raw gene expression data is normalized in the following steps:

1. Division by gene-wise standard deviation for scale normalization.
2. Quantile normalization across samples to reduce global expression biases.
3. Log-transformation: $x \mapsto \log(1 + x)$ to stabilize variance and suppress outliers.
4. For stress response studies, log fold-changes are computed:

$$\Delta_{\text{stress}} = \log(1 + x_{\text{stress}}) - \log(1 + x_{\text{control}})$$

This effectively makes each stress related axis show the fold change in gene expression under stress, e.g. “drought in root divided by control root”.

4 Construction of Expression Vector Space

- Biological replicates belonging to the same axis (tissue, stressed tissue, etc.) are averaged post-normalization.
- Each gene is represented as a vector $\vec{p} \in \mathbb{R}^n$, where n is the number of conditions or tissues. Hence the analysis is embedded in an n -dimensional ambient space.
- Expression vectors are stored in Fortran arrays. *Importantly* Fortran is column major and vectors *must* be columns.
- Two K-D Trees are constructed:
 - A raw-space K-D Tree (unnormalized vectors) for Euclidean queries.
 - A normalized-space K-D Tree (unit-length vectors) for cosine similarity and angular queries.
- The first axis is designated the **Anchor Axis (AA)** for visualization.

4.1 Vector Space Storage in Fortran

All expression vectors in Tensor Omics are stored in a central memory structure optimized for performance and access in Fortran. This architecture ensures efficient storage, lookup, and downstream computation.

- **vec_store:** A two-dimensional Fortran array of size $n \times v$, where n is the number of expression dimensions (e.g., tissues or conditions), and v is the number of vectors (e.g., genes). Each column represents one expression vector.
- **cart_tree:** A K-D Tree constructed on the raw (unnormalized) vectors stored in **vec_store**. This structure enables rapid spatial neighborhood queries using Euclidean distance in the original expression space. This is not initialized until step 10.1.
- **sphere_tree:** A second K-D Tree constructed from unit-length normalized vectors derived from **vec_store**. This structure supports directional queries based on cosine similarity or angular distance. This is not initialized until step 10.1.

Together, these data structures allow both magnitude-sensitive and direction-sensitive operations across expression fields.

5 Gene Family Grouping and Centroid Estimation

- Gene families, populations, or functional groups are defined using metadata.
- Family centroids $\vec{o}$ are computed either as:
 - The mean vector of all family members.
 - The mean vector of a family’s orthologs only.
- All metadata and vectors can be serialized into **.txdata** binary files.

6 Space Interpolation and Smoothing

To support field-based analysis and ensure consistent representation across expression space, a regular structured grid is constructed. This grid forms the structural foundation for kernel-based interpolation and vector field smoothing.

6.1 Grid Step Size Estimation

The grid resolution is determined empirically from intra-family expression variation. Specifically, for each gene family, the Euclidean distance between each member’s expression vector and the corresponding family centroid is computed. These intra-family distances are aggregated across all families into a single distribution. The grid step size (e_g) is then set as the **first quartile** (25th percentile) of this distribution.

Note: Since this step requires family membership and centroid information, it must be performed after gene family grouping and centroid estimation (see Section 5).

6.2 Gaussian Kernel Interpolation

After grid construction, expression vectors are propagated onto the surrounding grid using a Gaussian kernel function centered at each grid vector’s position. This operation produces a smoothly interpolated vector field that approximates local expression structure in the semantic space.

For a given expression vector $\vec{v}$ and a grid point $\vec{g}$, the interpolation weight is computed as:

$$w(\vec{v}, \vec{g}) = \exp\left(-\frac{\|\vec{v} - \vec{g}\|^2}{\sigma^2}\right), \text{ with } \sigma = e_g, \text{ the grid-step-size}$$

Only expression vectors located within a cutoff radius of $R = 3\sigma$ from the grid point contribute to its interpolated value. This follows standard statistical convention: for Gaussian kernels, 99.7% of the total mass lies within three standard deviations. This choice ensures inclusion of nearly all local signal while excluding remote vectors that would introduce unrelated semantic influences.

The kernel width σ is set equal to the estimated grid step size, aligning spatial influence with the empirically defined resolution of expression variability. Each grid point $\vec{g}$ receives the kernel-weighted average of all expression vectors within its support radius:

$$\vec{v}_{\text{interp}}(\vec{g}) = \frac{\sum_{\vec{v} \in \mathcal{V}_g} w(\vec{v}, \vec{g}) \cdot \vec{v}}{\sum_{\vec{v} \in \mathcal{V}_g} w(\vec{v}, \vec{g})}$$

where $\mathcal{V}_g$ is the set of all original expression vectors within range of $\vec{g}$. No interpolated vectors are used in this step; only original gene expression vectors contribute.

6.3 Gaussian Kernel Smoothing

After interpolation, a secondary smoothing step is applied to reinforce field coherence. This step uses the same Gaussian kernel formulation but is conceptually and operationally distinct from interpolation.

In smoothing, both the interpolated grid vectors $\vec{v}_{\text{interp}}$ and the original expression vectors $\vec{v}_{\text{orig}}$ contribute to the smoothed result. However, smoothing is applied only to the interpolated grid points, and the smoothed vectors are *not* recursively used in further smoothing steps. This design ensures mathematical clarity, avoids progressive drift or oversmoothing, and enables efficient parallel computation.

The smoothed vector at grid point $\vec{g}$ is computed as:

$$\vec{v}_{\text{smooth}}(\vec{g}) = \frac{\sum_{\vec{v} \in \mathcal{V}_g \cup \mathcal{G}_g} w(\vec{v}, \vec{g}) \cdot \vec{v}}{\sum_{\vec{v} \in \mathcal{V}_g \cup \mathcal{G}_g} w(\vec{v}, \vec{g})}$$

where:

- $\mathcal{V}_g$ are original expression vectors within $R = 3\sigma$ of $\vec{g}$,
- $\mathcal{G}_g$ are neighboring interpolated grid vectors (e.g., in a Moore neighborhood),
- All weights are computed using the same Gaussian kernel with width σ .

Importantly, smoothed vectors are not used to further smooth neighboring grid points. This avoids recursive blending and ensures that the smoothing operation can be executed independently across grid points, supporting massive parallelization on CPU or GPU backends.

Clarification regarding mathematical consistency: This two-step method preserves both the semantic geometry and the quantitative scale of the original expression space. Interpolation is based solely on raw expression vectors, and smoothing includes original anchors, thereby preventing topological drift. The choice of $R = 3\sigma$, combined with symmetric Gaussian kernels and fixed kernel width, ensures that no bias is introduced into the vector field and that all operations remain invariant under rotation and translation in expression space.

Interpretation: Interpolation constructs a continuous field over grid coordinates. Smoothing reinforces continuity and prepares the field for robust downstream operations such as flow estimation, rotation analysis, or module surface detection.

7 Empirical Distribution Estimation

To enable robust outlier detection and hypothesis-free exploration, Tensor Omics constructs empirical distributions of core metrics across all gene families.

7.1 Relative Distance Index (RDI)

For each gene family, we define the centroid $\vec{o}$ as the mean of all ortholog vectors $\vec{o}_i$ across species. To assess the magnitude of a gene’s divergence from this conserved expression center, we compute the Euclidean distance:

$$d = \|\vec{p} - \vec{o}\|$$

where $\vec{p}$ is the expression vector of the gene being evaluated.

To normalize this distance and enable scale-independent comparison across families, we compute a family-specific scaling factor as the maximum distance between any ortholog vector and the ortholog centroid:

$$d_{\text{scale}} = \max_i \|\vec{o}_i - \vec{o}\|$$

This normalization is linear in the number of orthologs and avoids the quadratic cost of pairwise distance computation. The Relative Distance Index (RDI) is then defined as:

$$\text{RDI} = \frac{\|\vec{p} - \vec{o}\|}{d_{\text{scale}}}$$

In cases where ortholog annotations are unavailable or the family contains only a single ortholog, a statistical smoothing strategy is used to estimate an appropriate scaling factor. Specifically, for each gene family, we calculate:

$$\text{FS}_i = \text{median}_j \left(\|\vec{f}_j - \vec{o}_i\| \right) \quad \text{and} \quad \text{FD}_i = \text{sd}_j \left(\|\vec{f}_j - \vec{o}_i\| \right)$$

where $\vec{f}_j$ are the expression vectors of all family members and $\vec{o}_i$ is the family centroid (mean of all $\vec{f}_j$). A smoothing function $f(\cdot)$, e.g. via LOESS, is fit to the set $\{(\text{FS}_i, \text{FD}_i)\}$ across all families. This yields an empirical mapping from median distance to expected spread. For families lacking orthologs, the smoothed value $f(\text{FS})$ is used as the denominator in:

$$\text{RDI}_{\text{smoothed}} = \frac{\|\vec{p} - \vec{o}\|}{f(\text{FS})}$$

This hybrid strategy ensures that RDI remains well-defined, interpretable, and comparable across evolutionary, functional, and statistical use cases in Tensor Omics.

All RDI values are pooled across all families to build an empirical distribution of relative intra-family divergence. This enables percentile-based filtering of significant shifts (e.g., 95th percentile for outlier detection) without requiring parametric assumptions.

Note: While RDI values are used to identify significant divergence, the unnormalized distances d_i are retained for all downstream geometric computations, such as vector field construction and shift vector analysis.

7.2 Relative Angle Index (RAI)

The angular difference between a vector $\vec{p}$ and its centroid $\vec{o}$ is computed using the cosine formula:

$$\text{RAI} = \cos^{-1} \left(\frac{\vec{p} \cdot \vec{o}}{\|\vec{p}\| \cdot \|\vec{o}\|} \right)$$

This identifies divergence in direction. Thresholds (e.g. 95%) are derived from the pooled distribution of angles.

7.3 Relative Axis Signal (RAS)

For each unit-normalized expression vector $\vec{p}^{\text{norm}}$, we consider its component $\vec{p}_i^{\text{norm}}$ along each axis i (e.g., tissue). For each axis, the 5th percentile across all vectors is computed as the inactivity threshold. A gene is considered neofunctionalized in axis i if:

$$\vec{o}_i^{\text{norm}} < T_i^{(5\%)} \quad \text{and} \quad \vec{p}_i^{\text{norm}} > T_i^{(5\%)}$$

where $\vec{o}^{\text{norm}}$ is the centroid.

7.4 Rank-normalized Signal Index (RSI)

To assess the local signal strength of expression or shift vectors across the space, Tensor Omics introduces the *Rank-normalized Signal Index (RSI)*. This index provides a scale-independent estimate of the absolute magnitude of a vector, contextualized within the empirical distribution of magnitudes in the current field.

Given a set of vectors

$$\mathcal{V} = \{\vec{v}_1, \dots, \vec{v}_N\} \subset \mathbb{R}^n$$

each vector's Euclidean norm is computed:

$$s_i = \|\vec{v}_i\|$$

These norms are assembled into a global magnitude distribution. To reduce the impact of noise and outliers, we extract the empirical 5th and 95th percentiles:

$$Q_5 = \text{Percentile}_{5\%}(\{s_i\}), \quad Q_{95} = \text{Percentile}_{95\%}(\{s_i\})$$

For each vector, the RSI is then computed as a clipped, percentile-normalized score:

$$\text{RSI}_i = \min \left(1, \max \left(0, \frac{s_i - Q_5}{Q_{95} - Q_5} \right) \right)$$

This formulation maps raw magnitudes into the unit interval $[0, 1]$, where values near 0 indicate weak or noise-like vectors, and values near 1 indicate vectors with strong signal relative to the field. Clipping ensures robustness even in the presence of heavy-tailed or multimodal distributions.

Note: RSI applies uniformly to expression fields and shift fields. In the context of signal-based subfield detection, it provides a direct and empirically grounded method for excluding noise vectors and prioritizing high-signal regions. In future extensions, RSI may also be computed on scalar fields derived from second-order tensors (e.g., Frobenius norm of surface tensors).

7.5 Relative Frobenius Index (RFI)

For any set of d vectors $\{\vec{v}_1, \dots, \vec{v}_n\}$, e.g. gene families, populations, or genes belonging to the same biological process or molecular function group, the vectors are assembled as columns in a matrix $M \in \mathbb{R}^{n \times m}$. The signal strength of this vector-group (matrix) is computed as:

$$\text{RFI} = \frac{\|M\|_F}{\sqrt{m}} = \frac{\sqrt{\sum_{i=1}^n \sum_{j=1}^m M_{ij}^2}}{\sqrt{m}}$$

This provides a normalized estimate of signal per vector and is used to assess subfield strength.

Note: The Frobenius norm is here used as a scalar estimate of the total signal energy contributed by a set of expression vectors. The matrix $M \in \mathbb{R}^{n \times m}$ contains m gene vectors (columns) embedded in a fixed n -dimensional semantic space. The RFI, defined as $\|M\|_F / \sqrt{m}$, reflects the average per-vector magnitude in a root-mean-square sense. This formulation emulates the cumulative effect of a gene set on a test particle exposed to their additive influence, without requiring square or self-adjoint structure. It supports comparison of sets with arbitrary cardinality m , all embedded in a shared space $\mathbb{R}^n$.

All empirical distributions are stored in Fortran arrays and used to assign significance thresholds for downstream classification and visualization.

8 Outlier Detection and Shift Vector Calculation

After constructing the vector space and identifying gene families, Tensor Omics detects divergent expression profiles within each family by comparing individual expression vectors to their respective family centroid.

Let $\vec{o} \in \mathbb{R}^n$ denote the centroid of a given family (typically the mean ortholog expression vector), and let $\vec{p}$ denote the expression vector of a candidate paralog. The *raw shift distance* between $\vec{p}$ and the centroid is defined as:

$$d_{po} = \|\vec{p} - \vec{o}\|$$

To determine whether this shift is significant, we normalize the raw distance using the maximum intra-family distance from the centroid. Let $\vec{f}_i$ denote all other expression vectors in the same family. The *Relative Distance Index (RDI)* is calculated as:

$$\text{RDI}_{po} = \frac{d_{po}}{\max_i \|\vec{f}_i - \vec{o}\|}$$

Vectors with an RDI greater than the 95th percentile of the empirical RDI distribution (pooled across all families) are flagged as significant outliers.

Important: Only the raw shift distance d_{po} is used for downstream geometric interpretation, such as vector field construction, trajectory analysis, and clock plot projection. The normalized RDI is used solely for filtering purposes, to robustly identify biologically meaningful divergence across gene families.

Identified outliers are stored as index references in a dedicated Fortran array. For each significant outlier, a corresponding *shift vector* is calculated:

$$\vec{s}_{po} = \vec{p} - \vec{o}$$

These shift vectors are added to the central vector space store ('vec_store') and indexed in the relevant K-D Trees (see section 4.1).

9 Tissue Versatility Calculation

Cosine of the angle θ between an expression vector and the space diagonal constant ($\vec{d}$):

$$\cos(\theta) = \frac{\vec{p} \cdot \vec{d}}{\|\vec{p}\| \cdot \|\vec{d}\|}$$

where $\vec{d} = (1, 1, \dots, 1)$ is the space diagonal and a constant.

This measure informs about a gene expression tissue (axis) specificity, or versatility, respectively. The larger θ the more the closer an expression vector's localization to one of the space's axes is. Thus its expression is more specific to the tissue represented by that axis. Conversely, the smaller θ the closer the expression vectors localization to tissue uniform expression along the space diagonal $\vec{d}$ and thus the more versatile the gene's expression in the tissues (axes).

10 Clock Plot Projection and Rotation Angle Computation

- All shift vectors are projected into the **Relative Axes Plane (RAP)**, orthogonal to the space diagonal and intersecting the origin.
- The following quantities are calculated for each outlier:
 - Projected shift vector in RAP.
 - Normalized (unit-length) projection.
 - Clock hand vector from origin to projected point.
 - Rotation angle (standardized as clockwise) between projected centroid and outlier.
- Results are stored in a structured Fortran matrix **rap_store**.
 - For each outlier i , store:
 - * Rows 1 to n : $\vec{s}_i^{\text{RAP}}$
 - * Rows $n + 1$ to $2n$: $\vec{s}^{\text{RAP, norm}}$
 - * Rows $2n + 1$ to $3n$: $\vec{s}_i^{\text{clock}}$
 - * Row $3n + 1$: rotation angle in radians

10.1 Efficient Vector Lookup Using K-D Trees

To enable fast and scalable querying of gene expression vectors, Tensor Omics employs K-D Trees—an efficient data structure for multidimensional nearest-neighbor searches. The trees are constructed at the very end of the pipeline. Four trees in total, two for all expression and shift vectors, where both are contained in **vec_store** (see 4), and two further for the RAP vectors in **rap_store** (see 10).

Two distinct trees are constructed:

- **cart_tree**: A K-D Tree built from the raw (unnormalized) expression vectors. It supports neighborhood searches in the original Euclidean space, useful for detecting local density patterns or spatial coherence among genes.
- **sphere_tree**: A second K-D Tree constructed from unit-length normalized vectors. This tree enables fast lookup of vectors with similar direction, allowing angular queries based on cosine similarity.

These lookup structures are only built during the final preprocessing stage, prior to interactive analysis. This deferred construction minimizes computational overhead during earlier pipeline steps and ensures consistency with the final state of the vector space. Together, the two trees provide complementary geometric access to the expression landscape.

11 Geometric Tools for Semantic Interpretation

Tensor Omics provides a suite of geometric tools—referred to as *semantic geometry measures*—that support interpretation, subfield detection, and comparative analysis of expression vector fields. These tools operate directly on the vector space and are designed to align with intuitive biological hypotheses.

11.1 Direction-Based Subfield Collector

To detect expression subfields aligned with a biologically meaningful trend, the user defines a direction-of-interest (DOI) vector $\vec{v}_{\text{doi}} \in \mathbb{R}^n$. This may correspond to, for example, increased healing rate, higher oil content, or a linear combination of expression axes.

For each expression vector $\vec{v}_i$, the cosine similarity with the DOI is computed. The method supports two modes, controlled by a Boolean input argument:

- **Normalization enabled:** both vectors are normalized to unit length, and the cosine is computed as:

$$\cos(\theta_i) = \vec{v}_i^{\text{norm}} \cdot \vec{v}_{\text{doi}}^{\text{norm}}$$

This mode emphasizes *direction* over magnitude and is suited for analyses where only the semantic trend matters, such as expression coordination in transcription factors or between tissues.

- **Normalization disabled:** the cosine is computed relative to the magnitude of the expression vector:

$$\cos(\theta_i) = \frac{\vec{v}_i \cdot \vec{v}_{\text{doi}}^{\text{norm}}}{\|\vec{v}_i\|}$$

This mode retains the influence of vector magnitude and is recommended when working in fold-change spaces (e.g., stress responses), where high expression shifts carry biological meaning.

The resulting cosine values $\cos(\theta_i)$ are used to construct an empirical distribution. Vectors with the strongest projection onto the DOI (e.g., top 5% by cosine value) are selected as a directional subfield. These can then be characterized using the Subfield Descriptor (Section 11.3).

11.1.1 When to Normalize (Direction-Only Interpretation)

Normalization is recommended when:

- The biological question relates to the **direction of change** in expression across conditions.
- Expression magnitude is not meaningful (e.g., low-expression transcription factors).
- Comparing semantic shifts across gene families or tissues.

11.1.2 When Not to Normalize (Magnitude Matters)

Omitting normalization is appropriate when:

- The space encodes **magnitude changes**, e.g., fold-change after stress.
- High-magnitude responses are biologically meaningful, such as upregulation in immune response genes.
- The DOI is used to detect strong response contributors in a specific context.

11.1.3 Selection of Directional Subfield

The cosine values $\cos(\theta_i)$ are collected across all vectors and an empirical distribution is formed. The top 5% (or another percentile threshold) are selected as the directional subfield—those vectors with the strongest alignment (largest projection) onto the DOI.

The selected subset is returned as a list of vector indices and can be further analyzed using subfield descriptors (Section 11.3).

11.2 Signal-Based Subfield Detection

In addition to metadata-driven and direction-of-interest-based methods for subfield collection, Tensor Omics supports a purely signal-based approach for identifying coherent expression or shift regions. This method—referred to as **Signal-Based Subfield Detection**—operates solely on vector magnitude and can be applied to both expression and shift fields. It provides a simple yet robust means of discovering compact, high-activity modules without assumptions about directionality or coherence, and serves as an efficient precursor to tensor field analysis.

11.2.1 Vector Collection

The procedure begins by selecting the tensor field to be analyzed:

- the **expression field**, consisting of gene expression vectors $\vec{p} \in \mathbb{R}^n$, or
- the **shift field**, with shift vectors defined as $\vec{s} = \vec{p} - \vec{o}$, where $\vec{o}$ is the gene family’s centroid.

The full set of vectors is assembled as:

$$\mathcal{V} = \{\vec{v}_1, \dots, \vec{v}_N\} \subset \mathbb{R}^n$$

Each $\vec{v}_i$ is selected from one or more of the following:

- Original expression or shift vectors (raw),
- Interpolated vectors at grid points,
- Smoothed vectors obtained from post-interpolation Gaussian smoothing.

For each vector $\vec{v}_i$, the signal strength measure depends on the chosen field:

- In the **expression field**, signal strength is evaluated using the Rank-normalized Signal Index (RSI), as described in Section 7.4. This ensures percentile-based normalization across diverse signal ranges.
- In the **shift field**, signal strength is computed directly via the Euclidean norm $\|\vec{s}_i\|$, which reflects absolute expression change relative to the centroid.

This modular design enables application across multiple resolution levels, from raw data to interpolated or smoothed fields. While the default implementation targets interpolated shift vectors, the framework generalizes to any well-defined vector field, including future fields derived from second-order tensors.

Note: In future extensions, this method may be applied to scalar fields derived from surface or covariance tensors (e.g., Frobenius norm of second-order structures). In such cases, the same region detection logic can be applied to identify coherent high-curvature or high-rotation zones.

11.2.2 Magnitude Distribution and Noise Filtering

To isolate signal-dense regions, the Euclidean norms $\|\vec{v}_i\|$ of all vectors in $\mathcal{V}$ are computed. An empirical noise threshold is then derived by estimating the lower 5th percentile:

$$\tau = \text{Percentile}_{5\%} \left(\{\|\vec{v}_i\|\}_{i=1}^N \right)$$

Vectors with norm less than τ are considered sub-threshold and excluded from further region growth.

11.2.3 Region Growing

From each unmarked vector $\vec{v}_j$ with $\|\vec{v}_j\| \geq \tau$, a local subfield is grown radially:

- Initialize the region with the seed $\vec{v}_j$ at radius $r = 0$.
- Increment the radius stepwise, adding unmarked neighboring vectors within radius r , provided they meet the magnitude threshold τ .
- If a full radius expansion includes only sub-threshold vectors, the region terminates.

All spatial distances are computed in the native semantic space $\mathbb{R}^n$ using Euclidean distance. Grid vectors and original data vectors may both serve as candidates in the same collection, depending on analysis scope.

11.2.4 Region Marking and Continuation

All vectors added to a region are marked. The procedure continues iteratively over the remaining unmarked, non-noise vectors until no new regions can be seeded.

11.2.5 Interpretation

This method segments the vector field into coherent *signal-rich subfields* based solely on local vector magnitude. In the expression field, the resulting regions may reflect domains of elevated transcriptional activity. In the shift field, they may reveal adaptation zones, such as tissue-specific neofunctionalization or spatially localized responses to perturbation. Because no directionality or covariance modeling is involved, this method is resilient to noise and sample sparsity.

Note: The threshold τ may be adjusted (e.g., 1%, 10%) to modulate sensitivity. The procedure generalizes naturally to higher-order tensors by applying the same algorithm to scalar-valued summary fields (e.g., Frobenius norms of second-order tensors).

11.3 Subfield Descriptor

Each collected subfield is characterized by aggregating its expression vectors into a matrix $M \in \mathbb{R}^{n \times k}$, where each of the k columns is a gene expression vector embedded in the fixed n -dimensional semantic space. The structure of the subfield is quantified using the following derived quantities:

- **Flow direction:** The first eigenvector of the covariance matrix $C = MM^T$, representing the dominant direction of signal flow across the subfield.
- **Rotation vector:** The antisymmetric (skew-symmetric) component of the covariance matrix is defined as:

$$A = \text{skew}(C) := \frac{1}{2}(C - C^T)$$

This decomposition follows conventions from continuum mechanics and enables separation of directional flow and local rotational tendency. The antisymmetric matrix A encodes the local rotation axis and magnitude of net curl induced by the subfield’s geometry.

- **Signal strength:** The Frobenius norm $\|M\|_F$ measures the total signal energy contributed by all expression vectors in the subfield.
- **Directional spread:** The matrix rank of M , estimated via singular value decomposition, indicates the number of independent directions represented in the subfield. A rank of one implies collinearity; higher ranks indicate multi-directional heterogeneity.

To enable comparison across subfields of varying size k , the signal strength can be normalized by the root of the cardinality:

$$\text{RFI} = \frac{\|M\|_F}{\sqrt{k}}$$

This Relative Frobenius Index (RFI) provides a density-independent measure of average vector magnitude per subfield.

Note: All direction vectors $\vec{v} \in \mathbb{R}^n \setminus \{\vec{0}\}$ are assumed to be non-zero, as normalization and angular analysis are undefined for the zero vector. If an unscaled antisymmetric component $A = C - C^T$ is used instead of the factorized form, this corresponds to measuring the total (unweighted) rotation strength. The scaled form $\frac{1}{2}(C - C^T)$ is preferred for decomposition consistency and interpretability.

11.4 Workbench for Interactive Exploration

The Tensor Omics workbench is a lightweight, browser-based environment for exploring the expression space and its geometric structures. It supports:

- 2D and 3D projections (e.g., RAP plane, PCA), with zoom and rotation.
- Efficient filtering via metadata or geometric queries using K-D Trees.
- Visual encoding of vector properties (color, shape, size) by metadata attributes.
- Highlighting of gene families, pathways, or other annotated subsets.

Plots are exportable (e.g., to SVG or PDF), and the current view state can be serialized and restored for reproducibility e.g. by attaching it as request parameters and thus enabling linking to preconfigured interactive plots. This supports sharing, publication, and reproducible workflows.

12 Time Series Interpolation and Contextual Smoothing

Tensor Omics applies a two-stage process to prepare time series expression trajectories for vector field analysis. This approach addresses the common problem of sparsely sampled or unevenly spaced time points in transcriptomic time courses and ensures that divergence detection remains sensitive and robust across genes and families.

12.1 Stage 1: Per-Gene Trajectory Interpolation

For each gene, we construct an interpolated expression trajectory using Gaussian kernel-based interpolation. Let the original time points be $t_1, t_2, \dots, t_n$, with corresponding expression vectors $\vec{e}_{t_i} \in \mathbb{R}^d$. These are interpolated to a denser trajectory with $T = \max(10n, 30)$ total time points, equally spaced over the original time interval.

Interpolation is performed independently per gene, without considering other genes. This preserves the native trajectory shape while increasing temporal resolution. For each interpolated time point t , a Gaussian kernel $K(t, t_i)$ is applied over all original points:

$$K(t, t_i) = \exp\left(-\frac{(t - t_i)^2}{2\sigma_i^2}\right)$$

where σ_i is the local kernel width, chosen adaptively based on sampling density (see below).

12.2 Stage 2: Contextual Smoothing Across Gene Families

To reduce noise and incorporate contextual biological similarity, each interpolated trajectory is smoothed using a Gaussian kernel over time, now also borrowing from ortholog and paralog trajectories.

Each smoothed time point $\vec{e}'_t$ is computed as a weighted average of nearby time points from:

- The same gene (denoted as G)
- Orthologs within the family (denoted as O)
- Paralogs within the family (denoted as P)

The total weight assigned to each group is:

$$\text{Contextual Weights: } \frac{6}{9} \text{ for } G, \quad \frac{2}{9} \text{ for } O, \quad \frac{1}{9} \text{ for } P$$

Within each group, the weight per point is distributed evenly:

$$w_{g_i} = \frac{6}{9|G|}, \quad w_{o_i} = \frac{2}{9|O|}, \quad w_{p_i} = \frac{1}{9|P|}$$

Each point is also assigned a Gaussian kernel weight based on temporal distance from the target time point:

$$K(t, t_i) = \exp\left(-\frac{(t - t_i)^2}{2\sigma_i^2}\right)$$

The final smoothing weight for each point is the product of its contextual weight and kernel weight. The smoothed expression vector is computed as:

$$\vec{e}'_t = \frac{\sum_i w_i K(t, t_i) \vec{e}_{t_i}}{\sum_i w_i K(t, t_i)}$$

12.3 Adaptive Kernel Width

The Gaussian kernel width σ_i is determined locally using symmetric nearest-neighbor spacing:

$$\sigma_i = \begin{cases} t_2 - t_1 & \text{if } i = 1 \\ \frac{1}{2}(t_{i+1} - t_{i-1}) & \text{if } 1 < i < n \\ t_n - t_{n-1} & \text{if } i = n \end{cases}$$

This ensures that smoothing respects the local temporal resolution of the data and avoids arbitrary or global smoothing parameters.

12.4 Implementation Notes

All interpolation and smoothing operations are performed using precomputed Gaussian kernels and group weights. The process is applied prior to shift vector calculation, ensuring that all time-series trajectories are temporally aligned, smooth, and comparable across genes and gene families.

This approach stabilizes vector field construction for downstream clock plots, rotation angle computation, and time-dependent divergence analysis.